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Uncertainty driven theoretical kinetics studies for CH_3OH ignition: $HO_2 + CH_3OH$ and $O_2 + CH_3OH$

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Abstract

A combination of global uncertainty screening and *ab initio* theoretical chemical kinetics is used to iteratively improve the mechanism of Li et al. [3] for the ignition of methanol at high pressure. The initial application of the screening analysis indicates that the $CH_3OH + HO_2$ reaction dominates the uncertainty in the predicted ignition delay for stoichiometric CH_3OH combustion at 1100 K and 20 bar. The rate coefficients for both product channels ($CH_2OH + H_2O_2$ and $CH_3O + H_2O_2$) in this reaction are predicted with *ab initio* transition state theory employing barriers and rovibrational properties obtained at the CCSD(T)/CBS//CASPT2/cc-pvtz level. The estimated uncertainty in these predictions is a factor of 2. The second iteration of the screening analysis indicates that the $CH_3OH + O_2$ reaction next dominates the uncertainty in the ignition delay at high pressure. The associated rate coefficient is updated using a two transition state model that employs CCSD(T)/CBS//CASPT2/cc-pvtz properties for the tight transition state and direct CASPT2/aug-cc-pvdz based variable reaction coordinate transition state theory for the barrierless formation of the long-range $CH_2OH \ldots HO_2$ complex. The final predictions for the ignition delay are a factor of 4 greater than those with the original model and the width of the distributions of ignition delay relative to its peak value decreases by a factor of 3. Further reduction in the uncertainty will require more accurate predictions for the $CH_3OH + HO_2$ reaction and new predictions for the $HO_2 + HO_2$ reaction. The predictions for the $CH_3OH + HO_2 \rightarrow CH_2OH + H_2O_2$, $CH_3OH + HO_2 \rightarrow CH_3O + H_2O_2$, and $CH_3O\text{-}H + O_2 \rightarrow CH_2OH + HO_2$ rate constants are well represented over the 400–2500 K temperature range, by the expressions $3.78 \times 10^{-29} T^{5.06} \exp(-5140/T)$, $5.54 \times 10^{-26} T^{4.12} \exp(-8170/T)$ and $5.95 \times 10^{-19} T^{2.27} \exp(-21520/T)$ cm^3 $molecule^{-1}$ s^{-1}, respectively, where T is in K.
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1. Introduction

The development of chemical mechanisms for combustion involves the compilation of rate coefficients, thermochemistry, and transport properties for a set of species and interconverting reactions. The choice of species and reactions is

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designed to reproduce important aspects of the chemical behavior occurring in the combustion environment. Mechanisms are generally validated via comparisons of model predictions with experimental data. Discrepancies between model and experiment sometimes suggest the need to include additional reaction pathways, or alternatively indicate shortcomings in the input data. Sensitivity analyses give an indication of the response of a given observable to the underlying rate coefficients and/or thermochemistry, in many cases providing an importance ranking for the key parameters driving the output uncertainty. This ranking provides motivation for further experimental and/or theoretical studies of the elementary reaction kinetics for key reactions [1,2].

Unfortunately, much experimental validation data is limited to pressures of a few bar or lower. In contrast, practical combustors, such as internal combustion engines, typically operate at pressures >10 bar. Furthermore, local sensitivity analysis does not account for the range of uncertainty in the rate and thermochemical coefficients, which can be especially large in cases where a limited number of fundamental studies have been performed. Ideally, one would like to improve the rate estimates for those reactions that contribute most strongly to the overall uncertainty in the predicted combustion properties under the conditions of interest to a particular practical combustion environment.

In this article, we use a global screening method to determine the reactions that contribute most strongly to the uncertainty in a model for the ignition of stoichiometric CH_3OH/O_2 at a pressure of 20 bar. A key component of this analysis involves the incorporation of uncertainty factors for each of the rate coefficients. We then use *ab initio* transition state theory to predict the rate coefficient for the reaction with the largest global sensitivity coefficient. Iteration of this approach results in a revised model with reduced uncertainty in the predicted ignition delay. We focus on methanol in this initial illustration of the basic approach due to the relative simplicity of its mechanism and its role in the hierarchical development of models for the oxidation of oxygenated hydrocarbons. As the simplest alcohol, a detailed understanding of its mechanism is relevant to the development of accurate mechanisms for important biofuels such as butanol.

Li et al. have recently developed a comprehensive kinetic mechanism for the combustion of CO, CH_2O, and CH_3OH [3] that builds on the early work of Held and Dryer [4]. It was validated against a wide range of experimental data including shock tube ignition delays up to 4.7 atm, flow reactor speciation at pressures up to 20 atm, and laminar flame speeds at 1 atm. This mechanism provides an excellent starting point for exploring the effects of high pressure on the ignition of methanol and as such is used as our base mechanism.

In this article, we emphasize the *ab initio* transition state theory calculations for two key reactions and their effect on the predicted ignition delay at higher pressures. For brevity, we provide only an overview of the global uncertainty screening approach. Furthermore, we restrict our attention to one representative set of initial conditions. A more detailed presentation of the methodology for the global sensitivity and uncertainty analysis [5], and a wider-ranging application to ignition of CH_3OH under a range of initial conditions [6] will be provided in separate articles. Also note that our intention is not to validate the model via a comparison with experiment for a particular set of initial conditions, but rather to illustrate the basic approach of an iterative coupling of global uncertainty screening and theoretical elementary reaction kinetics as a means to increase the fidelity of extrapolations of a given model to higher pressures where validation data may not exist, but where practical devices are operating.

2. Theory

2.1. Sensitivity and uncertainty modeling

The most sensitive reaction steps of the methanol mechanism are identified using a global screening method. In this approach, a target function is defined that is sensitive to the chemistry occurring in the regime of interest. Here, we focus on the ignition phase of the combustion and, thus, we adopt the auto-ignition delay time, τ, as the target. We define $\tau(k_1,k_2,\ldots k_N)$ as the time required to achieve an elevated temperature ($T_{\text{ignition}} = 1800$ K) for a homogeneous mixture in a constant volume thermally isolated chamber. The values of the input parameters k_i are distributed as independent random variables over the uncertainty range Δk_i for each of the coefficients and are chosen here from a uniform logarithmic distribution. Here, the input parameters are taken as the rate coefficients with uncertainty ranges for each of the A factors chosen on the basis of the uncertainty estimates provided in the reviews of Baulch et al. [7,8] and of Tsang et al. [9,10] (cf. supplementary data).

A total of 20,000 values of $\boldsymbol{k} = (k_1,k_2,\ldots k_N)$ are selected in the uncertainty volume surrounding the nominal values $\boldsymbol{k}_0$ for the given temperature and pressure. To ascertain which k_i's most strongly influence τ, the data points are numerically fit to the expansion of orthogonal functions [11,12]

$$\tau(k_1,k_2,\ldots,k_N) = \tau_0 + \sum_i A_i(k_i) + \sum_{i\neq j} B_{i,j}(k_i,k_j) + \ldots,$$

a form that has been employed in the high-dimensional model representation method [13,14]. To compute the contribution of a given k_j to the variation of τ within the uncertainty range, the partial variance of τ with respect to k_j is computed using a single first order term, $\sigma_j^2 = \left\langle \left(A_j - \langle A_j \rangle\right)^2 \right\rangle_k$. The global sensitivity index is defined as the ratio of the partial to total variances, i.e., $S_j = \sigma_j^2 / \left\langle (\tau - \langle \tau \rangle)^2 \right\rangle$. If the coupling between k_j's is small within the output function τ, then the sum of S_j is near 1.

The larger values of S_j identify the reactions that most strongly influence τ, which are thus targeted for re-evaluation with more accurate theoretical and/or experimental methods. It is crucial to appreciate that, unlike the local sensitivity coefficients $s_j = \partial \ln \tau / \partial \ln k_j|_{k_0}$, the global indices S_j reflect both the variability of τ with k_j and the size of the uncertainty range. Thus, e.g., a reaction may have a high s_j but a low S_j if the rate coefficient is known very accurately. Indeed, the use of the S_j-criterion leads to the sensible result that there is no need to update a k_j that is already well known. The global indices also account for those parameters that may have a low local sensitivity at the accepted nominal value, but high sensitivities in other regions of the adopted range.

2.2. Ab initio transition state theory

The hydroperoxyl radical reacts with methanol via direct H abstractions at one of two abstraction sites:

$$CH_3OH + HO_2 \rightarrow CH_2OH + H_2O_2 \quad (1)$$

$$CH_3OH + HO_2 \rightarrow CH_3O + H_2O_2 \quad (2)$$

The rovibrational properties of the stationary points for these reactions were studied with multi-reference second order perturbation theory (CASPT2) [15,16] employing Dunning's correlation consistent polarized valence triple zeta (cc-pvtz) basis set [17]. These CASPT2 calculations employed a five-electron five-orbital (5e, 5o) active space consisting of the HO_2 radical orbital, the OH σ, σ^* pair and a localized CH σ, σ^* pair. Higher level energy estimates were obtained from coupled cluster calculations with perturbative corrections for the effect of triple excitations [CCSD(T)] [18] at the CASPT2/cc-pvtz geometries. The CCSD(T) calculations are extrapolated to the complete basis set (CBS) limit [19] from explicit aug-cc-pvtz and aug-cc-pvqz calculations and employ spin-restricted Hartree–Fock wavefunctions within the spin-unrestricted coupled cluster formalism. The frequencies, rotational constants, and barrier heights for reactions (1) and (2) are reported in Table 1. All of the electronic structure calculations were done with the MOLPRO program package [20].

The rate coefficients for reactions (1) and (2) were obtained from conventional transition state theory employing rigid-rotor harmonic-oscillator energy approximations for all but the torsional degrees of freedom [21]. The torsional minima were determined from a complete sampling of the torsional degrees of freedom. Hindered rotor corrections were obtained from one-dimensional

Table 1
Stationary point properties for the $CH_3OH + HO_2$ potential surface.

Stationary point	Energies[a] (kcal/mol)	Frequencies[b] (cm^{-1})	Rotational constants[b] (GHz)
$CH_3OH + HO_2$	0.00	312, 1074, 1106, 1186, 1391, 1494, 1519, 1530, 3008, 3102, 3186, 3867[c]	24.0, 24.9, 128.8[c]
		1221, 1454, 3687[d]	32.9, 34.8, 620.0[d]
$CH_2OH + HOOH$	8.26	411, 518, 1068, 1237, 1381, 1517, 3201, 3350, 3865[e]	26.2, 30.1, 194.3[e]
		376, 957, 1335, 1443, 3799, 3826[f]	25.8, 26.7, 302.0[f]
$CH_3O + HOOH$	17.68	805, 964, 1134, 1397, 1423, 1542, 2936, 3071, 3134[g]	27.9, 27.8, 158.8[g]
$CH_3OH + HO_2 \rightarrow CH_2OH + HOOH$	15.27	1674i, 120, 156, 259, 328, 507, 537, 611, 1047, 1108, 1165, 1213, 1302, 1397, 1409, 1494, 1561, 3095, 3222, 3770, 3804	3.4, 4.3, 13.7
$CH_3OH + HO_2 \rightarrow CH_3O + HOOH$	21.24	2981i, 90, 125, 177, 264, 398, 626, 978, 1009, 1043, 1118, 1192, 1398, 1403, 1412, 1460, 1522, 2993, 3069, 3139, 3787	3.4, 4.1, 14.7

[a] UCCSD(T)/CBS//CASPT2/cc-pvtz including zero-point correction.
[b] CASPT2/cc-pvtz.
[c] CH_3OH.
[d] HO_2.
[e] CH_2OH.
[f] HOOH.
[g] CH_3O.

fits to the torsional potentials employing Pitzer–Gwinn like approximations and the $I^{(2,3)}$ moments of inertia [22]. Asymmetric Eckart tunneling corrections were also included.

The reaction of CH_3OH with O_2 also proceeds via direct abstraction at one of the two abstraction sites. However, in this case the abstraction transition state lies lower in energy than the asymptotic product energies. Thus, the reaction first produces a van der Waals complex, which then proceeds on to bimolecular products via passage through a separate long-range transition state:

$$CH_3OH + O_2 \rightarrow CH_2OH \cdots HO_2 \rightarrow CH_2OH + HO_2 \quad (3)$$

$$CH_3OH + O_2 \rightarrow CH_3O \cdots HO_2 \rightarrow CH_3O + HO_2 \quad (4)$$

The methoxy radical is 9.4 kcal/mol higher in energy than the CH_2OH species. Thus, channel (4) has a much lower rate coefficient than channel (3), and is safely neglected.

The barrier height and rovibrational properties of the stationary points in reaction (3) are reported in Table 2 and were obtained at the CCSD(T)/CBS//CASPT2/cc-pvtz level as described above for reactions (1) and (2). However, for this reaction we employed a (12e, 9o) active space consisting of the (6e, 4o) π space of O_2, the (4e, 4o) single CH and OH σ, σ^* pairs, and the (2e, 1o) O lone pair of CH_3OH. Notably, the present prediction of 46.72 kcal/mol for the endothermicity of reaction (3) is in quantitative agreement with the current Active Thermochemical Tables (ATcT) value of Ruscic (46.66 ± 0.12 kcal/mol) [23].

The microcanonical J-resolved reactive flux, N_{EJ}^{inner}, for the $CH_3OH + O_2 \rightarrow CH_2OH \cdots HO_2$ reaction is obtained with the same transition state theory methods as employed for reactions (1) and (2). Meanwhile, the flux, N_{EJ}^{outer}, for the $CH_2OH \cdots HO_2 \rightarrow CH_2OH + HO_2$ step is obtained with direct variable reaction coordinate transition state theory (VRC-TST) [24,25]. The latter calculations employ CASPT2(2e,2o)/aug-cc-pvdz energies where the two active orbitals are simply the radical orbitals of HO_2 and of CH_2OH. Center-of-mass dividing surfaces with separations of 3.4 Å and greater are considered in this direct VRC-TST analysis. The overall rate constant for reaction (3) is evaluated from the effective flux, N_{EJ}^{eff}, given by $1/N_{EJ}^{\text{eff}} = 1/N_{EJ}^{\text{inner}} + 1/N_{EJ}^{\text{outer}}$, which roughly accounts for the effect of the two transition states acting in series.

3. Results and discussion

The mechanism of Li et al. for CH_3OH combustion consists of thermochemical data for 18 species and rate coefficients for 93 reactions. The global first-order sensitivities for the ignition delay calculated with this mechanism are plotted in Fig. 1 for a stoichiometric mixture of CH_3OH and O_2 at an initial temperature of 1100 K and an initial pressure of 20 bar. The variance in the ignition delay is clearly dominated by the 79% contribution from $CH_3OH + HO_2$. This strong sensitivity arises from a large uncertainty of a fac-

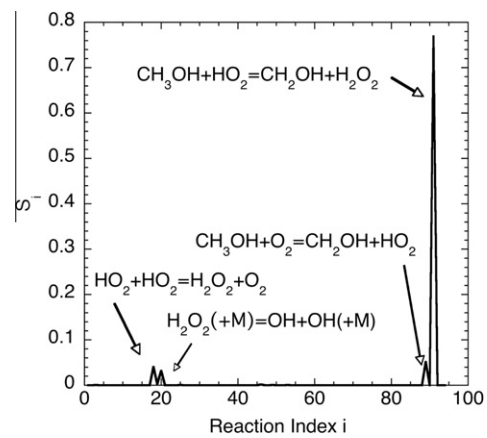

Fig. 1. Plot of the global first-order sensitivity indices for the ignition delay in methanol for the original mechanism of Li et al. [3].

Table 2
Stationary point properties for the $CH_3OH + O_2$ potential surface.

Stationary point	Energies[a] (kcal/mol)	Frequencies[b] (cm^{-1})	Rotational constants[b] (GHz)
$CH_3OH + O_2$	0.00	For CH_3OH see Table 1 1628[c]	 43.2, 43.2[c]
$CH_2OH + HO_2$	46.72	For CH_2OH see Table 1 1117, 1426, 3722[d]	 31.6, 33.4, 616.6[d]
$CH_3OH + O_2 \rightarrow CH_2OH + HO_2$	42.50	1222i, 147, 197, 302, 473, 639, 720, 1038, 1097, 1166, 1253, 1415, 1449, 1516, 1601, 3114, 3269, 3606	4.5, 3.7, 15.8

[a] UCCSD(T)/CBS//CASPT2/cc-pvtz including zero-point correction.
[b] CASPT2/cc-pvtz.
[c] O_2.
[d] HO_2.

tor of 10 for this rate coefficient [10], coupled with the dominant importance of the HO_2 radical at high pressures. This reaction also appeared as a significant reaction in the study of Li et al. [3], but in a much more limited sense because no consideration was given to the relative uncertainties in the input rate coefficients, and due to the emphasis on lower pressures and higher temperatures. This parallels earlier work on CO/H_2 where the $CO + HO_2$ reaction was highlighted in the high pressure study of Mittal et al. [1], but had not been identified as important by earlier sensitivity studies performed at atmospheric pressure [26].

The large uncertainty for reaction (1) is due to the difficulty of performing direct measurements. At low temperatures, the rate coefficient is too low for the reaction to be observable. At higher temperatures, the HO_2 radical decomposes too rapidly. Apparently, the only "experimental" result for this rate coefficient comes from a simulation of concentration profiles arising from methanol oxidation in the 800–900 K regime [27].

The present CCSD(T)/CBS//CASPT2/cc-pvtz based *ab initio* TST predictions for the rate coefficients for reactions (1) and (2) are illustrated in Fig. 2. The contribution from channel (1) is seen to dominate over that from channel (2), as might be expected from the relative energies of the two products and their corresponding transition states. Included in Fig. 2 are the values employed in a few chemical models for CH_3OH combustion [3,28,29]. Comparison of the present predictions with the model values illustrates that the presumed uncertainty of a factor of 10 in this rate coefficient is appropriate. The models of Ref. [3] and [28] differ by more than a factor of 10 at 1000 K. The model of Miller is in better agreement, but even that model presumes a significantly different temperature dependence. (Note that the discrepancy between the present predictions and the modeling values is not a criticism of the modeling efforts. There simply was insufficient information to accurately estimate this rate coefficient.) Over the key 600–2000 K range our best estimate of the uncertainty factor is between 1.5 and 2.5. This uncertainty arises roughly equally from uncertainties in the predicted barrier heights (~1 kcal/mol) and uncertainties in the state counting procedures (~50%).

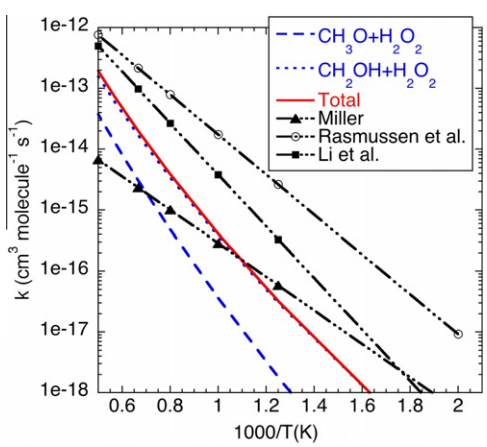

Fig. 2. Plot of the temperature dependent rate coefficients for the reaction of CH_3OH with HO_2. The dotted (blue), dashed (blue), and solid (red) lines denote the present predictions for the rate coefficients to produce $CH_2OH + H_2O_2$, $CH_3O + H_2O_2$, and the total products, respectively. The lines with symbols denote the rate coefficients for producing $CH_2OH + H_2O_2$ from the models of Refs. [3], [28], and [29].

There do not appear to have been any other theoretical studies of this reaction. We have recently demonstrated the value of theoretical predictions for the $CO + HO_2$ reaction, which is of similar importance in the high pressure ignition of H_2 and CO mixtures in a rapid compression machine [1,2]. Dean and coworkers [30,31] and Simmie and coworkers [32] have each predicted the kinetics of the related abstraction reactions of HO_2 with saturated alkanes. The factor of two agreement between theory and experiment for the calculations of Simmie and coworkers is in line with our estimated uncertainties. In contrast, the predictions of Dean and coworkers are up to a factor of five too fast, apparently due to inaccuracies in their more limited electronic structure calculations.

The global sensitivity indices are plotted in Fig. 3 for an updated mechanism that incorporates the present theoretical predictions for the rate coefficients for reactions (1) and (2). The sensitivity index for reaction (1) has decreased significantly due to the decrease in both the magnitude of the rate coefficient and its uncertainty. The greatest global sensitivity index is now for the

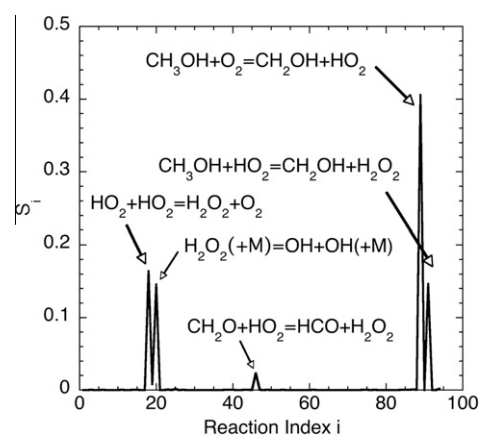

Fig. 3. Plot of the global first-order sensitivity indices for the ignition delay in methanol after updating the rate coefficients for the $CH_3OH + HO_2$ reactions.

$CH_3OH + O_2$ reaction. This is again due in part to the large uncertainty (a factor of 10) in the rate coefficient. Notably, the sensitivity index (94) for the newly included reaction (2) does not show up.

The present predictions for the $CH_3OH + O_2 \rightarrow CH_2OH + HO_2$ rate coefficient are plotted in Fig. 4. Again, the predictions, which have an uncertainty of about a factor of 2, are an order of magnitude lower than the values employed in the modeling studies of Refs. [3] and [28]. Notably, our recent related predictions for the analogous $CH_4 + O_2 \rightarrow CH_3 + HO_2$ reaction were in good agreement with the shock tube measurements of Srinivasan and Michael [33].

Incorporation of these predictions for the rate coefficient of reaction (3) yields the new sensitivity indices plotted in Fig. 5. At this point there are two reactions with nearly equal first-order sensitivities: $HO_2 + HO_2 \rightarrow H_2O_2 + O_2$ and again $CH_3OH + HO_2 \rightarrow CH_2OH + H_2O_2$. Furthermore, the H_2O_2 decomposition to $OH + OH$ has a similar sensitivity. We have already studied reaction (1) in the present analysis and so any further reductions in the overall uncertainty will require even more accurate estimates for this rate coefficient. The combination of higher level focal point schemes [34], with full treatments of the anharmonic effects, and a more accurate tunneling treatment may yield such a reduction. However, such calculations are complex and so were deemed beyond the scope of the present work. A detailed theoretical analysis for the $HO_2 + HO_2$ reaction would be worthwhile, since the values at high temperature are based on a single experimental study that extends up to only 1250 K and has a somewhat unusual temperature dependence [35]. Such a study is planned for future work. For the

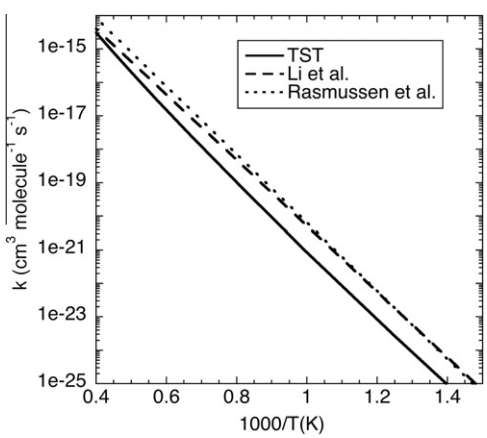

Fig. 4. Plot of the temperature dependent rate coefficients for the reaction of CH_3OH with O_2. The solid lines denote the present predictions, while the dashed and dot-dash lines denote the values employed in the modeling studies of Refs. [3] and [28], respectively.

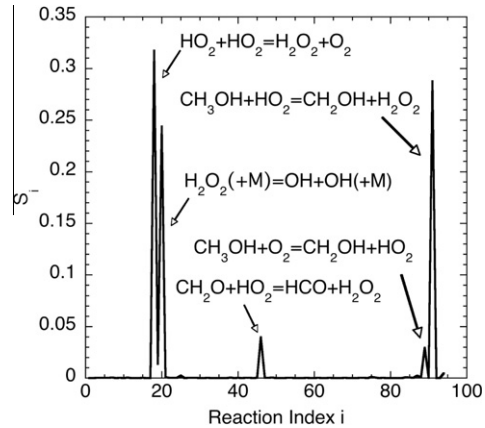

Fig. 5. Plot of the global first-order sensitivity indices for the ignition delay in methanol after updating the rate coefficients for the $CH_3OH + HO_2$ reactions and the $CH_3OH + O_2 \rightarrow CH_2OH + HO_2$ reaction.

H_2O_2 decomposition, we note that Miller and coworkers have recently provided what should be very accurate theoretical predictions [36].

Figure 6 illustrates the distribution of predicted ignition delays at different stages in the analysis. The theoretically predicted rate coefficients are each smaller than those employed in the original mechanism. Thus, the ignition delay increases as the mechanism is updated. For example, the nominal ignition delays are 0.11, 0.34, and 0.40 ms for the initial mechanism and after the first and second updates, respectively. The peaks in the distribution of predicted ignition delays are at smaller values of 0.033, 0.28, and 0.38 ms due in part to the non-linear dependence of the ignition delay on the rate coefficients. The full-width at half-maximum of the distributions scaled

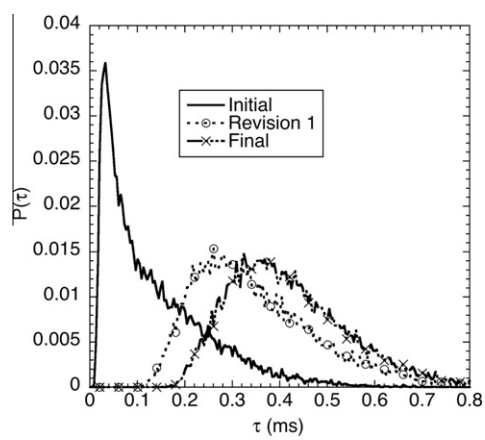

Fig. 6. Plot of the probability of a given ignition delay for the combustion of CH_3OH for the 3 stages of the mechanism.

by the average ignition delay decreases from 2.0 to 0.95 to 0.70 as the mechanism is updated, demonstrating the decrease in uncertainty of the predictions with updates.

4. Conclusions

The present iterative coupling of global sensitivity and uncertainty analysis with theoretical chemical kinetics yields an effective procedure for reducing the uncertainty in predicting ignition delays for CH_3OH at high pressure. This general approach should be widely applicable. This study also further demonstrates the importance of HO_2 reactions when considering high pressure and low temperature conditions.

Acknowledgments

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.proci.2010.05.066.

References

[1] G. Mittal, C.J. Sung, M. Fairweather, A.S. Tomlin, J.F. Griffiths, K.J. Hughes, *Proc. Comb. Inst.* 31 (2007) 419–427.
[2] X. You, H. Wang, E. Goos, C.-J. Sung, S.J. Klippenstein, , *J. Phys. Chem. A* 111 (2007) 4031–4042.
[3] J. Li, Z. Zhao, A. Kazakov, M. Chaos, F.L. Dryer, *Int. J. Chem. Kinet.* 39 (2007) 109–136.
[4] T.J. Held, F.L. Dryer, *Int. J. Chem. Kinet.* 30 (1998) 805–830.
[5] M.J. Davis, R.T. Skodje, A.S. Tomlin, in preparation.
[6] R.T. Skodje, A.S. Tomlin, S.J. Klippenstein, L.B. Harding, M.J. Davis, *J. Phys. Chem. A* (2010), doi:10.1021/jp1047002.
[7] D.L. Baulch, C.J. Cobos, R.A. Cox, et al., *Chem. Ref. Data* 21 (1992) 411–734.
[8] D.L. Baulch, C.J. Cobos, R.A. Cox, et al., *J. Phys. Chem. Ref. Data* 23 (1994) 847–1033.
[9] W. Tsang, W.F. Hampson, *J. Phys. Chem. Ref. Data* 15 (1986) 1087–1279.
[10] W. Tsang, *J. Phys. Chem. Ref. Data* 16 (1987) 471–508.
[11] A. Saltelli, M. Ratto, T. Andres, et al., *Global Sensitivity Analysis,The Primer*, John Wiley, 2008.
[12] I.M. Sobol', *Math Model Comput. Ex.* 1 (1993) 407.
[13] H. Rabitz, O.F. Alis, J. Shorter, K. Shim, *Comput. Phys. Commun.* 117 (1999) 11.
[14] T. Ziehn, A.S. Tomlin, *Environ. Model. Soft.* 24 (2009) 775.
[15] K. Andersson, P.-A. Malmqvist, B.O. Roos, *J. Chem. Phys.* 96 (1992) 1218.
[16] H.-J. Werner, *Mol. Phys.* 89 (1996) 645;
P. Celani, H.-J. Werner, *J. Chem. Phys.* 112 (2000) 5546.
[17] R.A. Kendall, T.H. Dunning Jr., R.J. Harrison, *J. Chem. Phys.* 96 (1992) 6796.
[18] K. Raghavachari, G.W. Trucks, J.A. Pople, M. Head-Gordon, *Chem. Phys. Lett.* 157 (1989) 479.
[19] J.M.L. Martin, O. Uzan, *Chem. Phys. Lett.* 282 (1998) 16.
[20] H.-J. Werner, P.J. Knowles, R. Lindh, et al., MOLPRO, version 2008.1, a package of ab initio programs, see http://www.molpro.ne.
[21] D.G. Truhlar, B.C. Garrett, S.J. Klippenstein, *J. Phys. Chem.* 31 (1996) 12771–12800.
[22] A.L.L. East, L. Radom, *J. Chem. Phys.* 106 (1997) 6655–6674.
[23] B. Ruscic, private communication, ATcT results based on the Core (Argonne) Thermochemical Network ver. 1.078 (2009).
[24] S.J. Klippenstein, *J. Chem. Phys.* 96 (1992) 367–371.
[25] L.B. Harding, Y. Georgievskii, S.J. Klippenstein, *J. Phys. Chem. A* 109 (2005) 4646–4656.
[26] I.Gy. Zsely, J. Zador, T. Turanyi, *Proc. Comb. Inst.* 30 (2005) 1273–1281.
[27] M. Cathonnet, J.C. Boetnner, H. James, *J. Chim. Phys.* 79 (1982) 475–478.
[28] C.L. Rasmussen, K.H. Wassard, K. Dam-Johansen, P. Glarborg, *Int. J. Chem. Kinet.* 40 (2008) 423–441.
[29] J.A. Miller, private communication.
[30] H.-H. Carstensen, A.M. Dean, O. Deutschmann, *Proc. Comb. Inst.* 31 (2007) 149–157.
[31] H.-H. Carstensen, A.M. Dean, *Proc. Comb. Inst.* 30 (2005) 995–1003.
[32] J. Aguilera-Iparraguirre, H.J. Curran, W. Klopper, J.M. Simmie, *J. Phys.Chem. A* 112 (2008) 7047–7054.
[33] N.K. Srinivasan, J.V. Michael, L.B. Harding, S.J. Klippenstein, *Comb. Flame* 149 (2007) 104–111.
[34] S.E. Wheeler, K.A. Robertson, W.D. Allen, H.F. Schaefer, Y.J. Bombie, J.F. Stanton, *J. Phys. Chem. A* 111 (2007) 3819–3830.
[35] H. Hippler, J. Troe, J. Willner, *J. Chem. Phys.* 93 (1990) 1755.
[36] S.R. Sellevag, Y. Georgievskii, J.A. Miller, *J. Phys. Chem. A* 133 (2009) 4457–4467.